

Machine Learning Lab(CS3066)

Lab Assignment- 1

1. Create a NumPy array of integers from 1 to 10 and find the mean and standard deviation of the array.
2. Write a python program to find the inverse of a square matrix using NumPy.
3. Using Matplotlib, plot a line graph for the function $y=x^2$ where x ranges from -10 to 10.
4. Using Pandas, create a DataFrame with the following data:

Name	Age	City
Alice	24	New York
Bob	30	Chicago
Charlie	29	San Francisco
David Putra	35	New York City

Filter the DataFrame to show only the rows where the age is greater than 28.

5. Write a python program to calculate the Determinant of a Matrix Using NumPy. Generate random data for x and y using NumPy (50 points each) and create a scatter plot using Matplotlib. Use different colors and markers for points in two ranges of x: $x < 0.5$ and $x \geq 0.5$.
6. Generate a random dataset of 50 points using NumPy, create a scatter plot of these points using Matplotlib, and calculate the mean of the x and y coordinates using Pandas.
7. Load the following dataset into a Pandas DataFrame:

Product	Price	Quantity
A	50	10
B	30	5
C	40	8

Calculate the total revenue for each product and the overall revenue.